

Task 1: Google Dork

Use Google Search to find:

- 1. PDF files related to Cyber Security**
- 2. PPT files related to Ethical Hacking**
- 3. Cyber Security reports published online**

Examples:

- **filetype:pdf cyber security**
- **filetype:ppt ethical hacking**

Write down:

- **Search Query Used**
- **What Information You Found**

To use Google Dorking techniques to find publicly available cyber security resources such as PDF files, PPT files, and reports.

Step 1:Open Google Chrome or any web browser.

Step 2:Go to: <https://www.google.com>

Step 3: Use the following search queries:

- **Search Query 1:-filetype:pdf cyber security**
- **Search Query 2:-filetype:ppt ethical hacking**
- **Search Query 3:-cyber security report filetype:pdf**

Step 4:Open some search results and observe the documents available publicly.

Step 5:

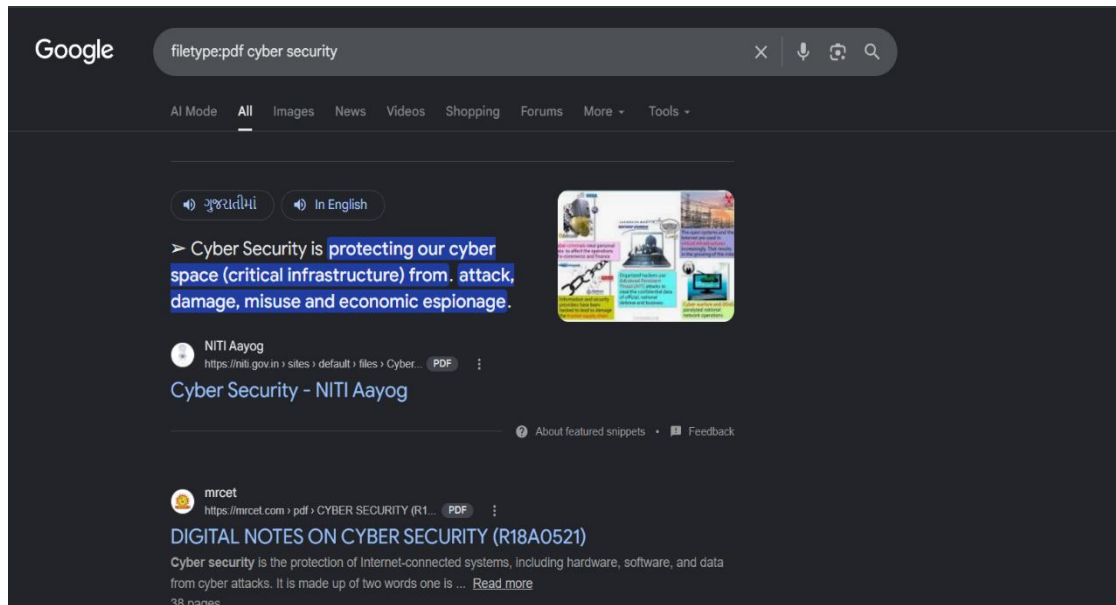
Take screenshots of:

- Search query
- Search results page
- Opened PDF/PPT document

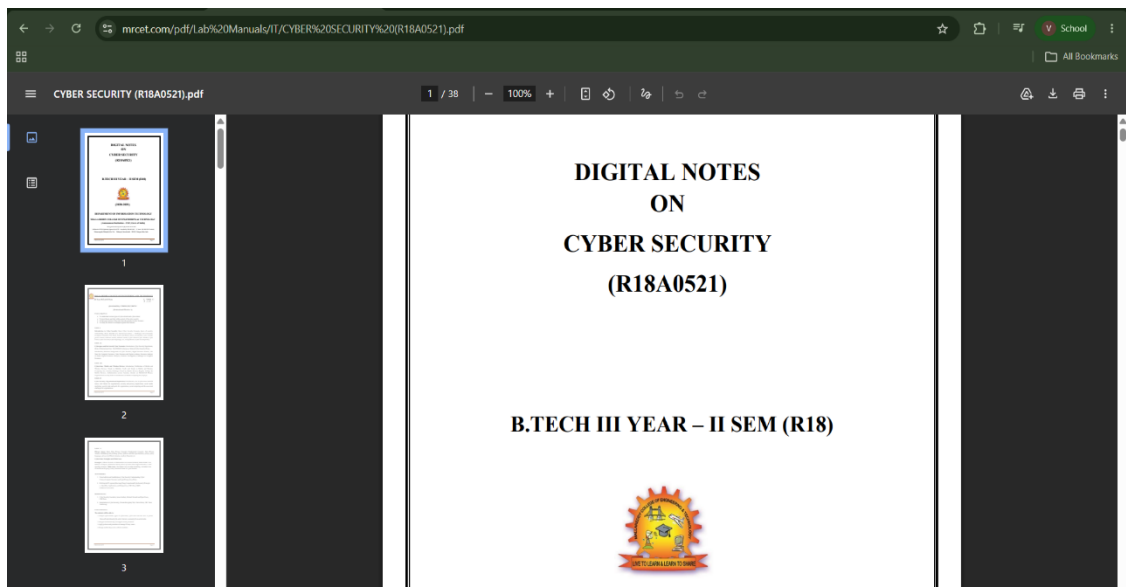
Screenshots:

- **Search Query 1:-**filetype:pdf cyber security

Searching the Query:



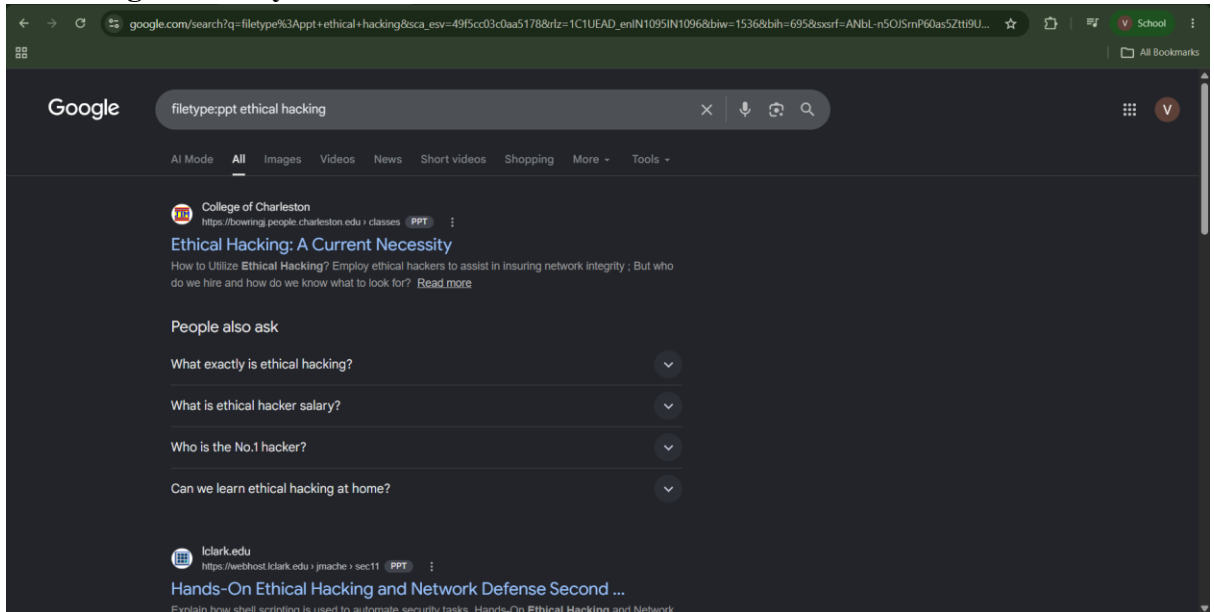
Downloaded pdf:



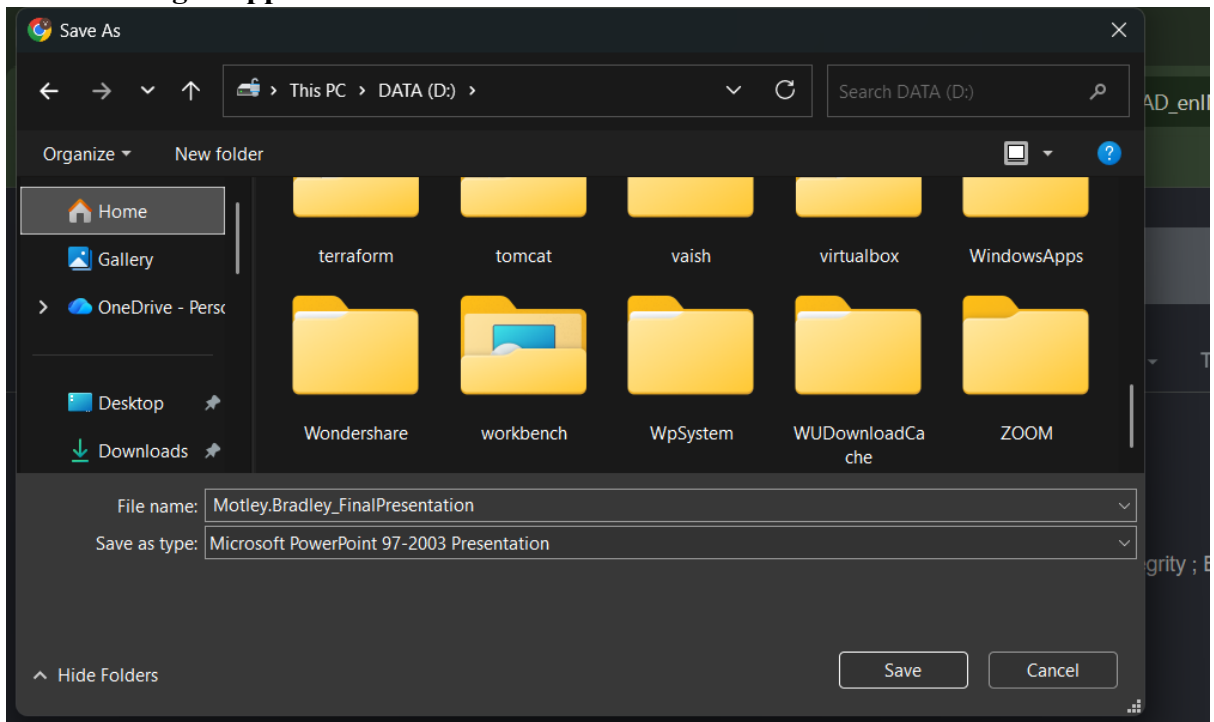
Information found: Found cyber security study materials, research papers, and security guidelines in PDF format.

- Search Query 2:-filetype:ppt ethical hacking

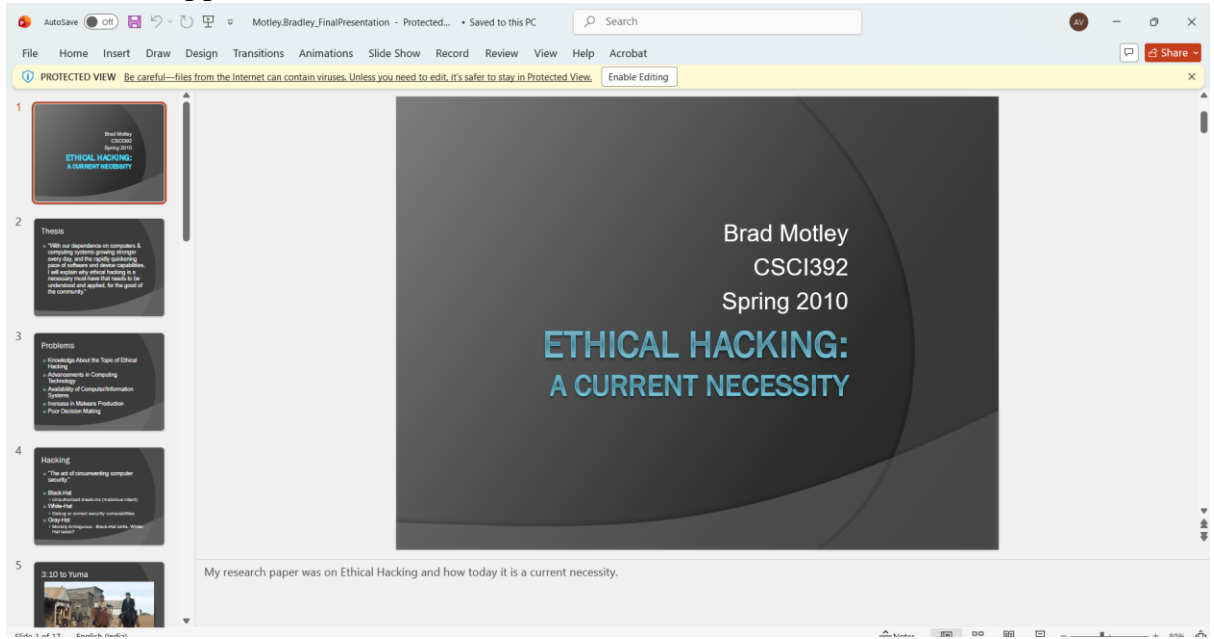
Searching the Query:



Downloading the ppt:



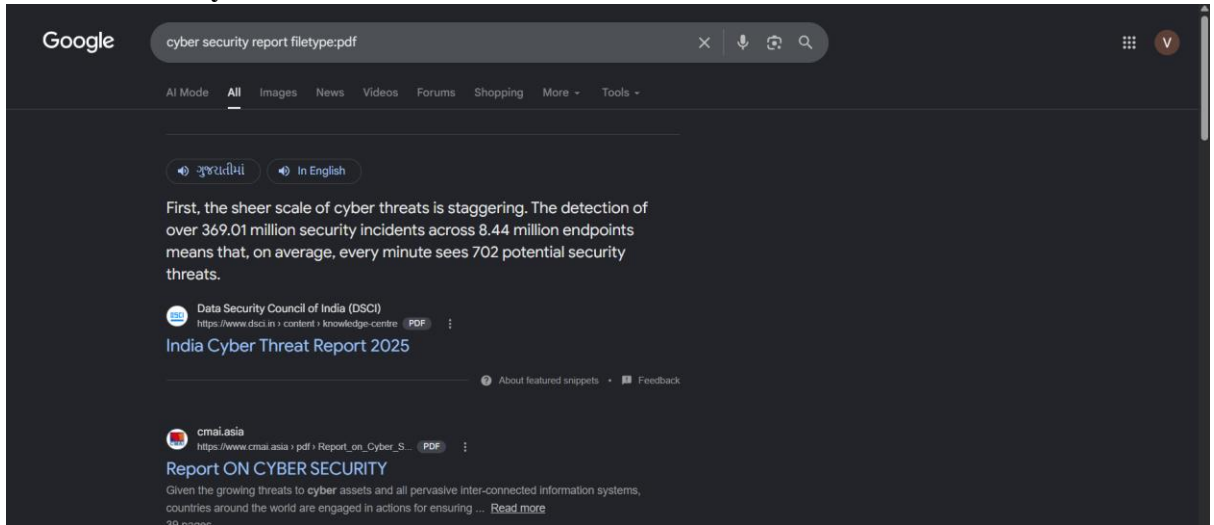
Downloaded ppt:



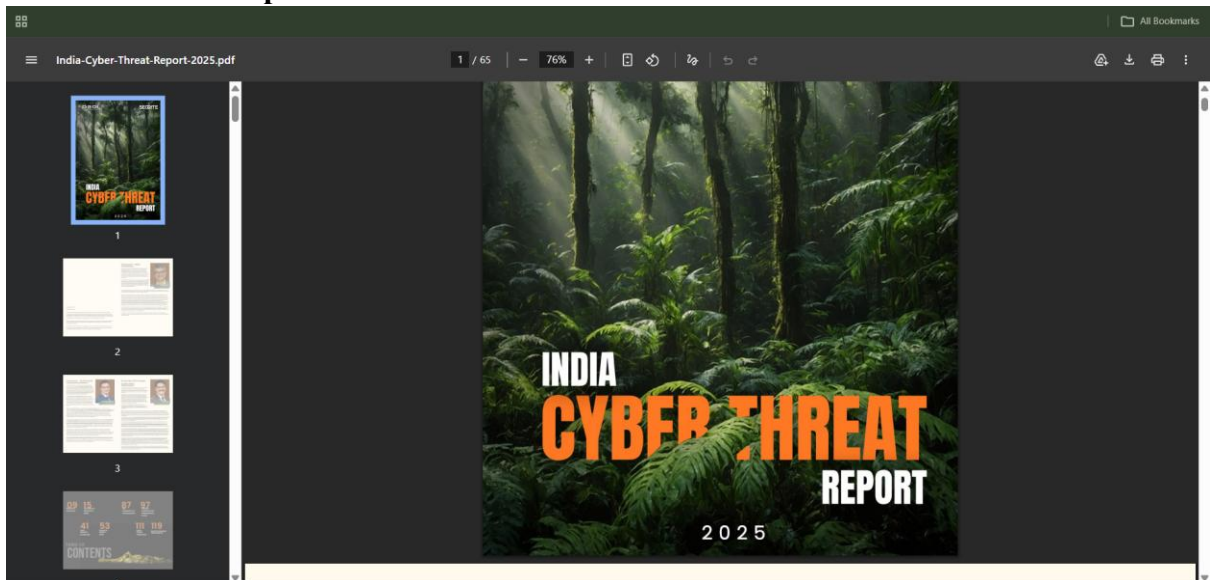
Information found: Found PowerPoint presentations related to ethical hacking concepts and penetration testing.

- **Search Query 3:-**cyber security report filetype:pdf

Searched Query:



Downloaded the report:



Information found: Found cyber security annual reports, threat reports, and vulnerability analysis reports.

Task 2: Website Technology Investigation

Visit any 5 websites.

Find:

1. Website Name
2. Technologies Used (Frontend, Backend if visible)
3. Is HTTPS enabled?
4. Is Login available?
5. Is Registration available?
6. What security features can you identify?

Website 1 – [Amazon](https://www.amazon.com)

Steps Performed

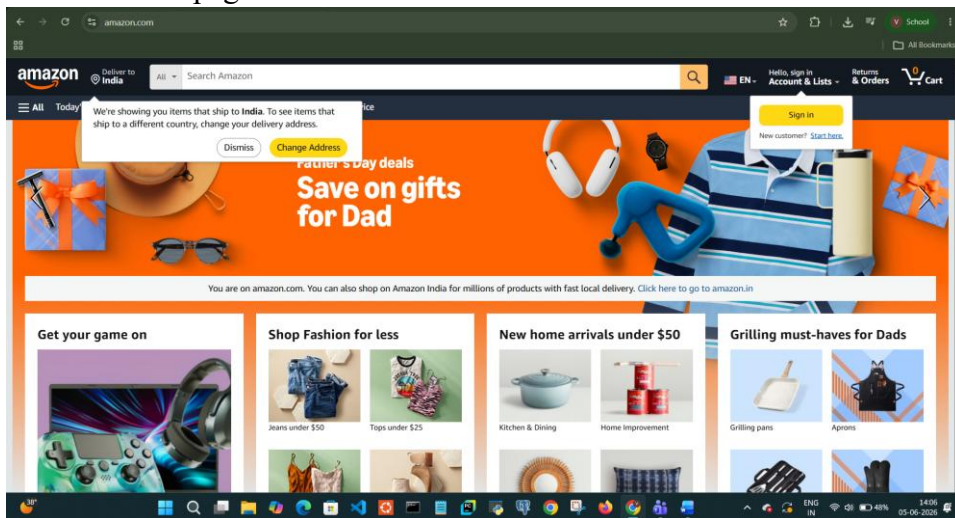
1. Opened Amazon website in Google Chrome.
2. Pressed **F12** to open Developer Tools.
3. Checked:
 - Network tab
 - Security tab
 - Sources tab
4. Observed HTTPS connection and login system.

Observations

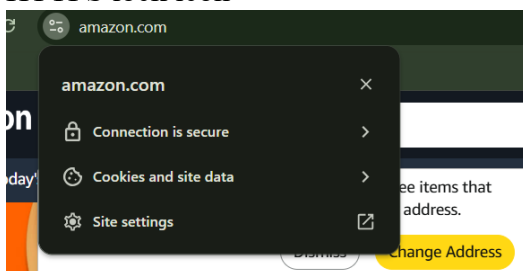
Parameter	Details
Website Name	Amazon
Frontend Technology	HTML, CSS, JavaScript, React
Backend Technology	Not directly visible
HTTPS Enabled	Yes
Login Available	Yes
Registration Available	Yes
Security Features Identified	HTTPS encryption, secure cookies, account authentication, login verification

Screenshots to Insert

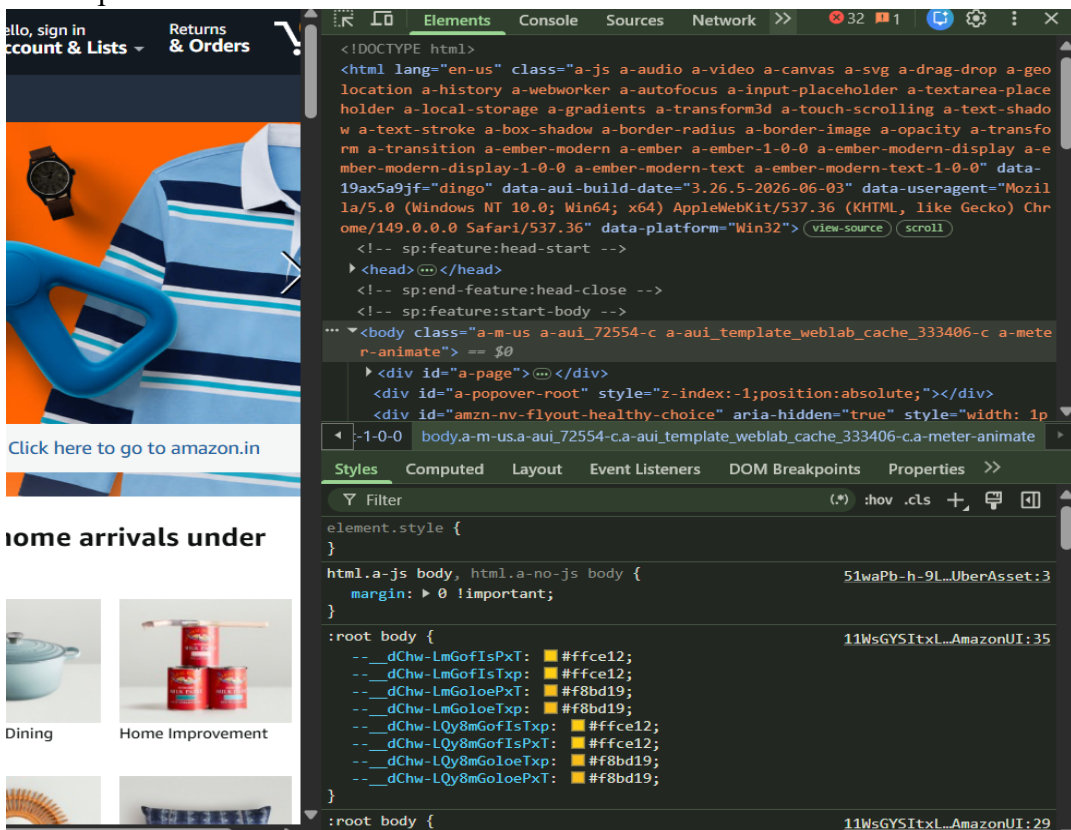
- Amazon homepage



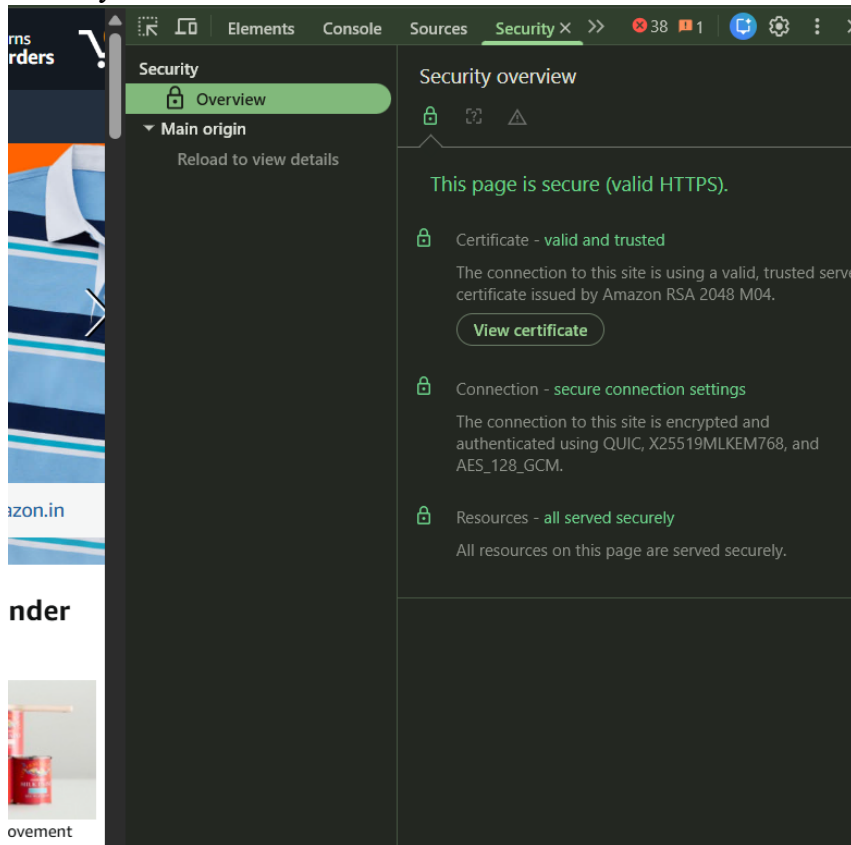
- HTTPS lock icon



- Developer Tools window



- Security tab



Website 2 – [Instagram](#)

Steps Performed

1. Opened Instagram website.
2. Pressed **F12**.
3. Inspected Developer Tools and Security tab.
4. Observed login and registration pages.

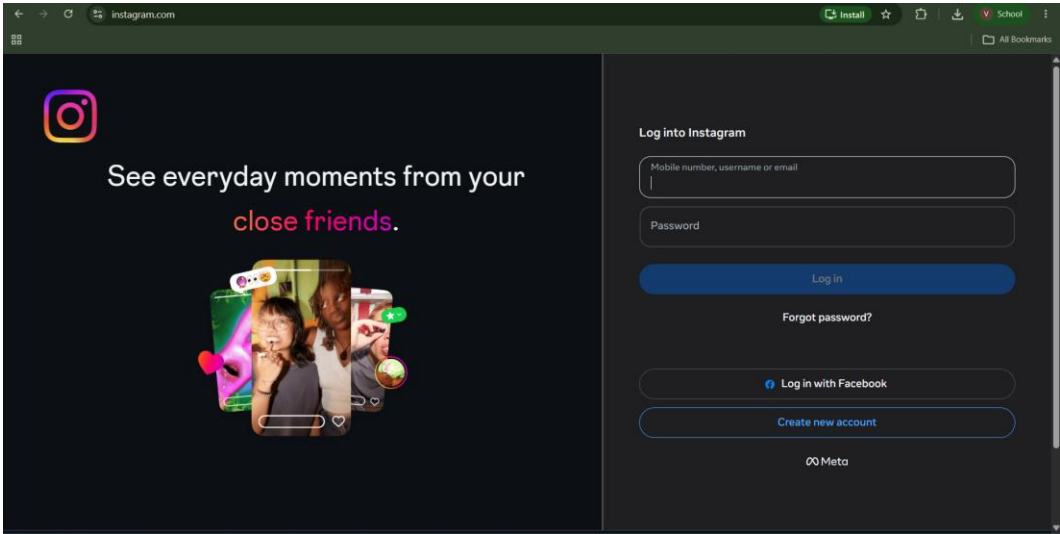
Observations

Parameter	Details
Website Name	Instagram
Frontend Technology	React, JavaScript
Backend Technology	Not directly visible
HTTPS Enabled	Yes

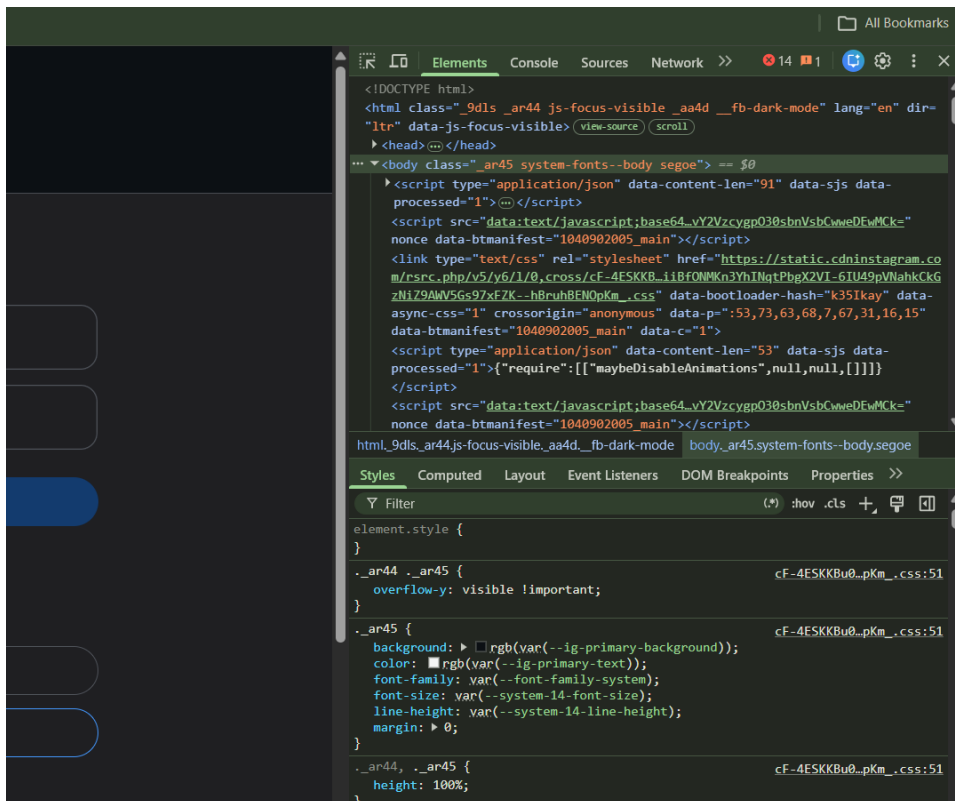
Parameter	Details
Login Available	Yes
Registration Available	Yes
Security Features Identified	HTTPS, secure login authentication, session management, secure cookies

Screenshots to Insert

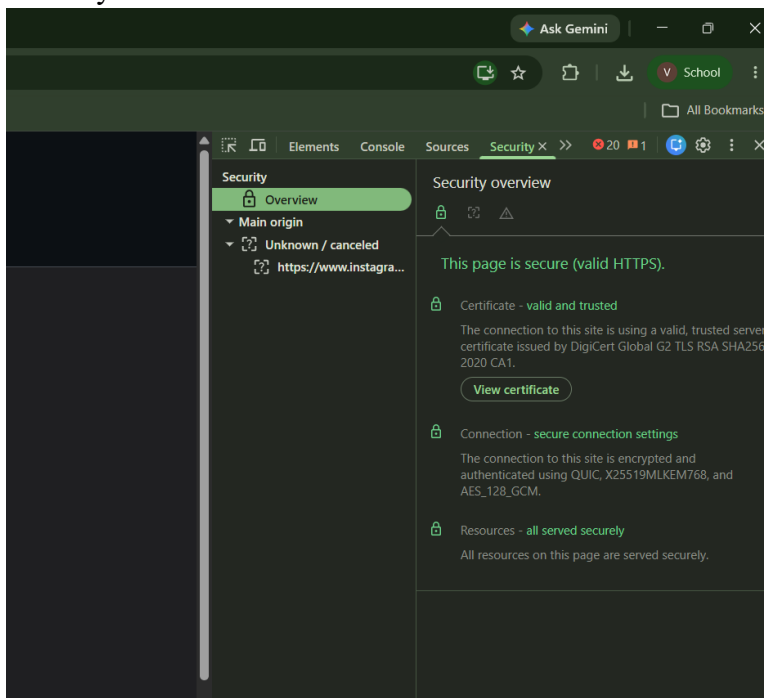
- Instagram homepage



- Developer Tools



- Security tab



Website 3 – YouTube

Steps Performed

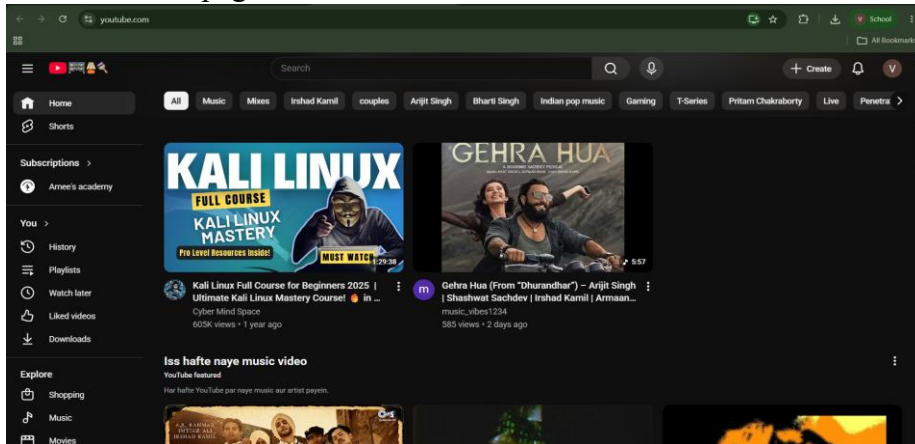
1. Opened YouTube website.
2. Opened Developer Tools using **F12**.
3. Checked network requests and security information.

Observations

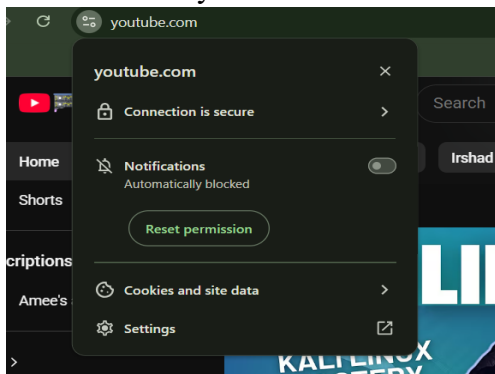
Parameter	Details
Website Name	YouTube
Frontend Technology	HTML, CSS, JavaScript, Polymer
Backend Technology	Not directly visible
HTTPS Enabled	Yes
Login Available	Yes
Registration Available	Yes
Security Features Identified	HTTPS, Google account authentication, encrypted communication

Screenshots to Insert

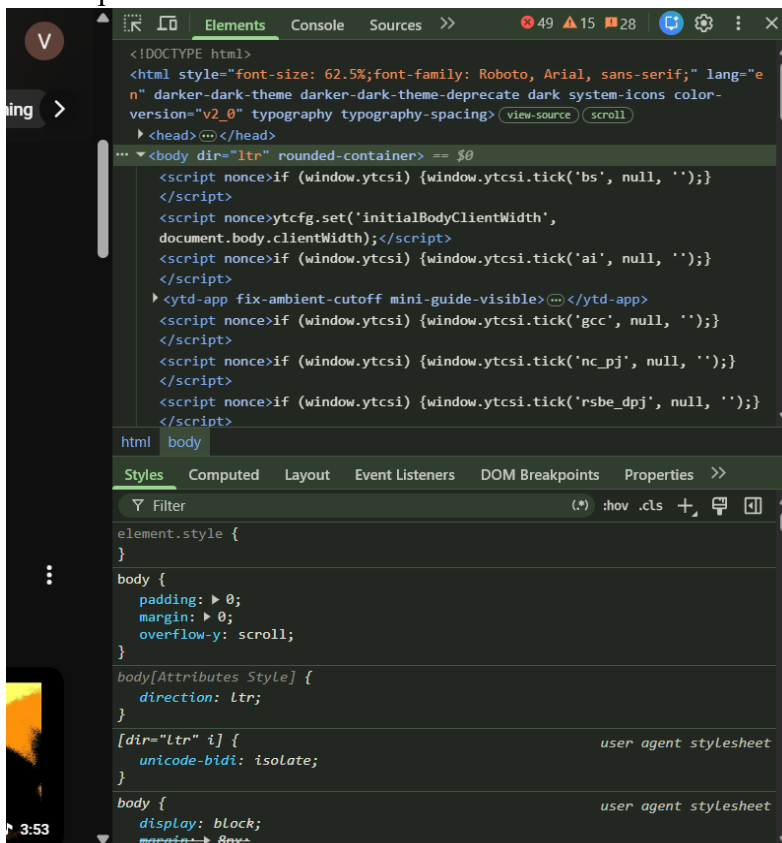
- YouTube homepage



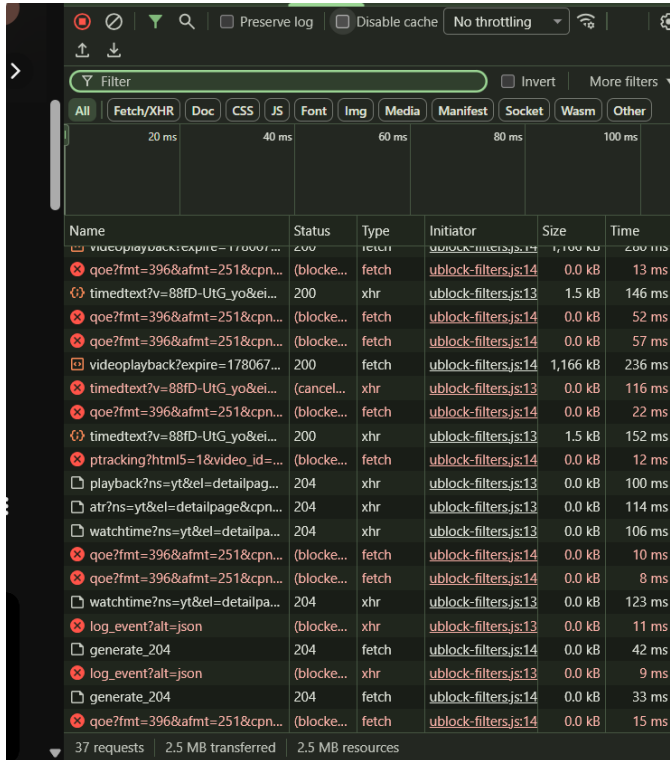
- HTTPS security icon



- Developer Tools



- Network tab



Website 4 – Netflix

Steps Performed

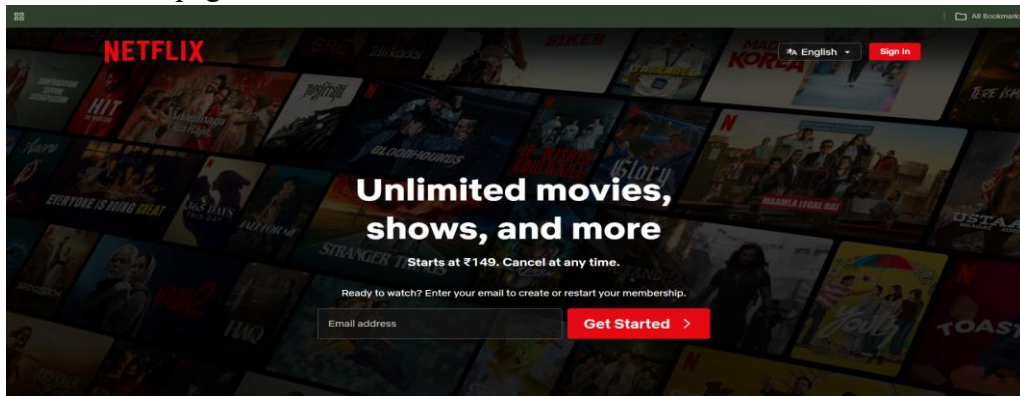
- 1. Opened Netflix website.
- 2. Pressed **F12** to open browser Developer Tools.
- 3. Inspected Security and Sources tabs.

Observations

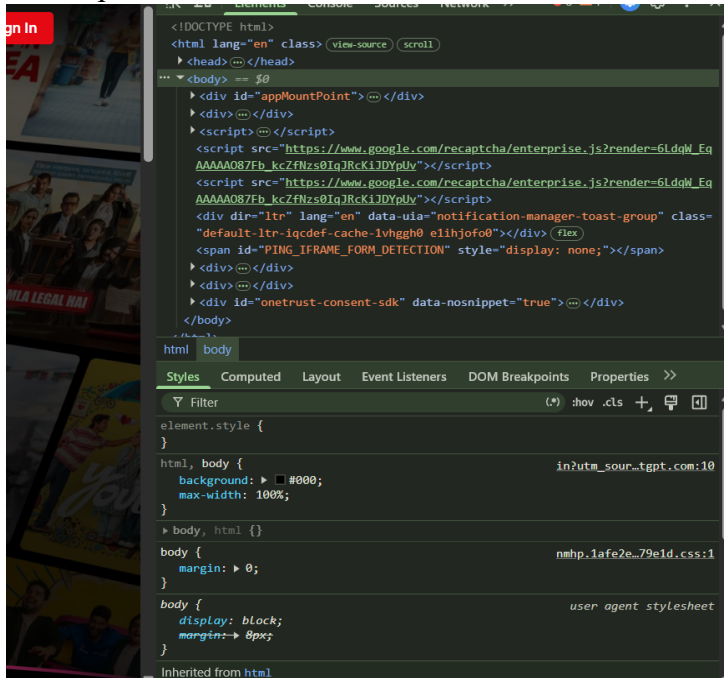
Parameter	Details
Website Name	Netflix
Frontend Technology	React, JavaScript
Backend Technology	Not directly visible
HTTPS Enabled	Yes
Login Available	Yes
Registration Available	Yes
Security Features Identified	HTTPS encryption, account authentication, secure session handling

Screenshots to Insert

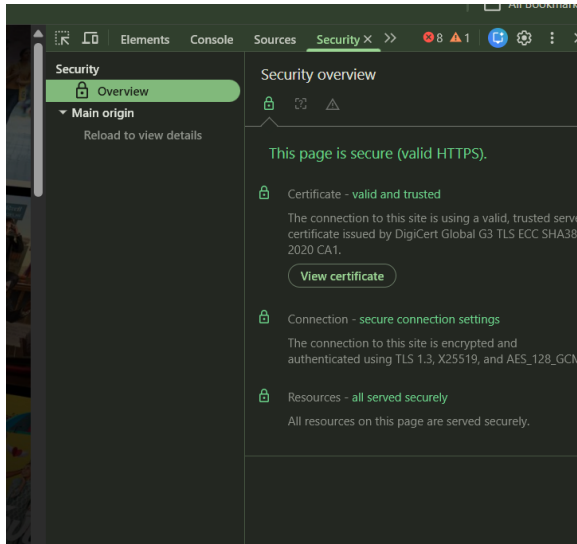
- Netflix homepage



- Developer Tools



- Security tab



Website 5 – Flipkart

Steps Performed

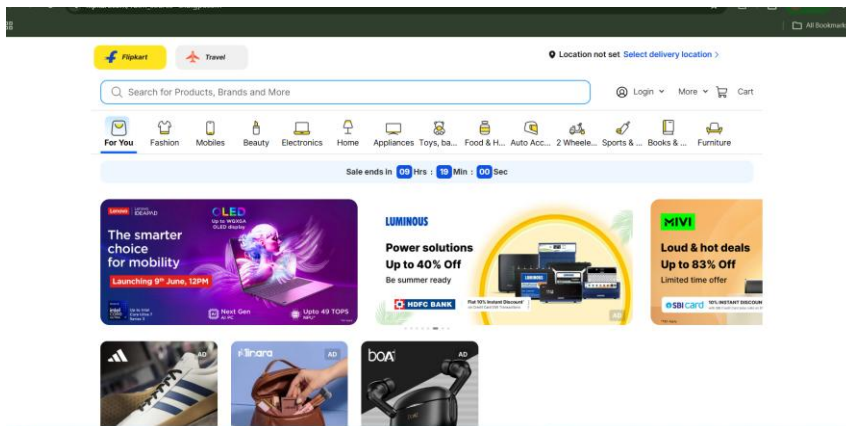
1. Opened Flipkart website.
2. Opened Developer Tools using **F12**.
3. Checked security information and login system.

Observations

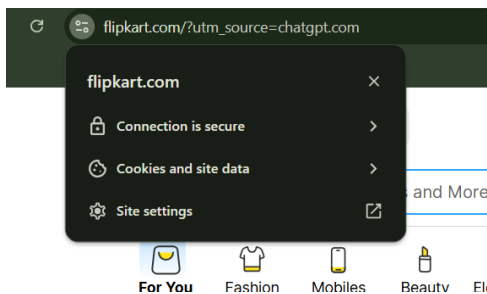
Parameter	Details
Website Name	Flipkart
Frontend Technology	HTML, CSS, JavaScript, React
Backend Technology	Not directly visible
HTTPS Enabled	Yes
Login Available	Yes
Registration Available	Yes
Security Features Identified	HTTPS, secure login authentication, encrypted communication

Screenshots to Insert

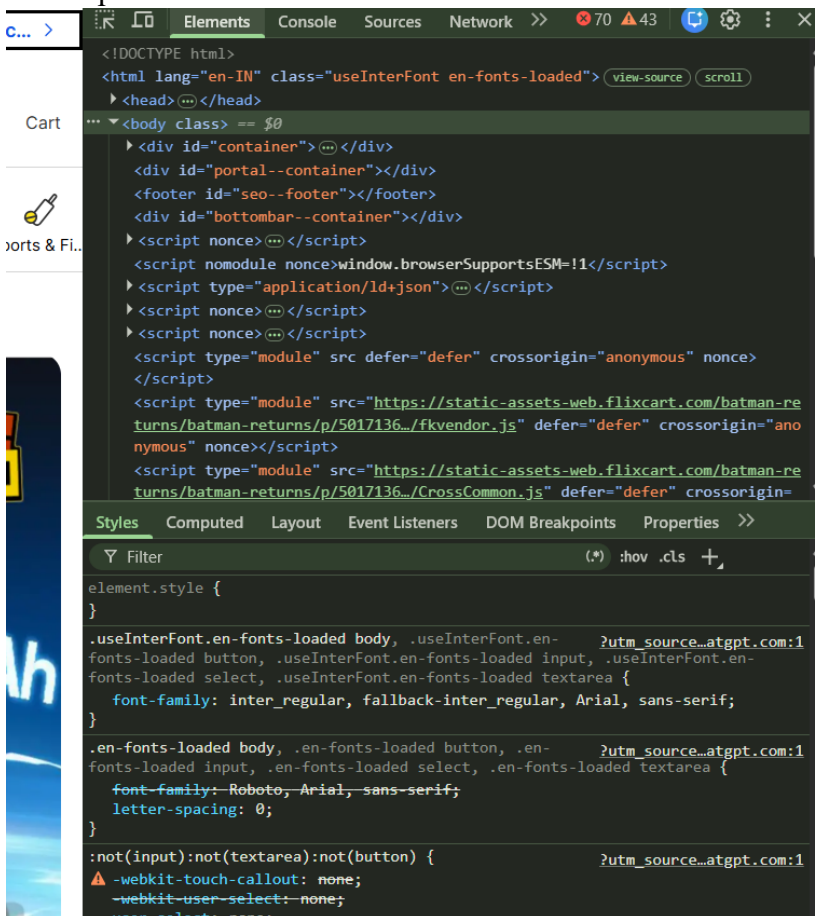
- Flipkart homepage



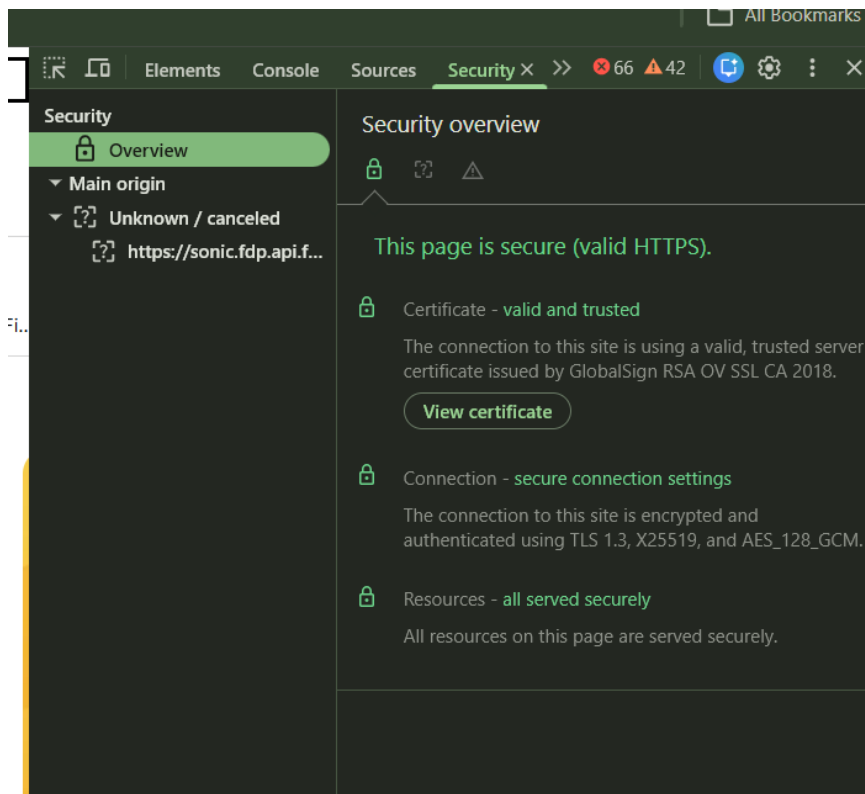
- HTTPS lock icon



- Developer Tools



- Security tab



Task 3: Email Header Investigation

Open an email in Gmail.

Click:

More → Show Original

Observe:

1. Sender Email
2. Return Path
3. Authentication Results
4. SPF
5. DKIM

Take screenshots and write what you found.

To analyze email headers and identify email authentication methods.

Step 1: Open Gmail.

Step 2: Open any received email.

Step 3: Click: More (three dots) → Show Original

Step 4: Observe:

- Sender Email

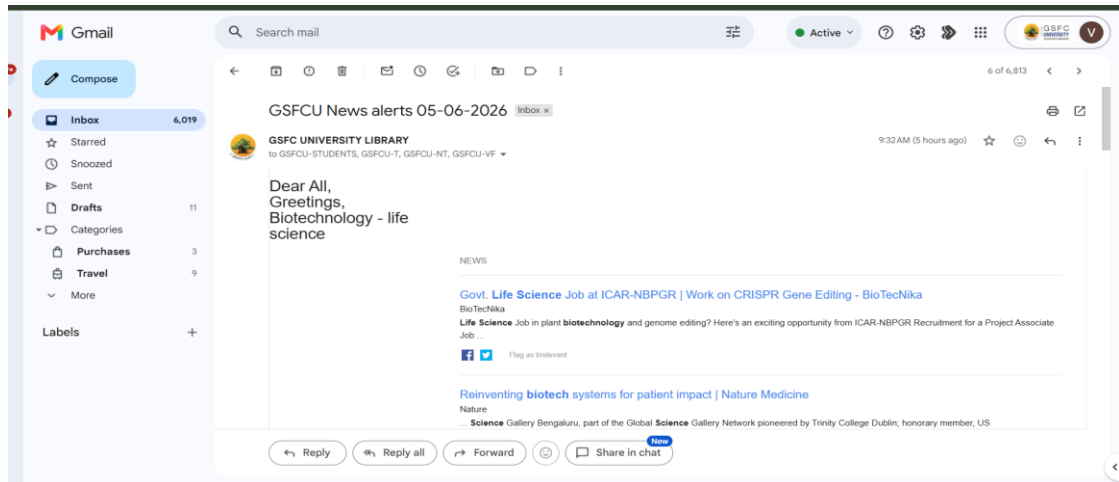
- Return Path
- SPF
- DKIM
- Authentication Results

Observations

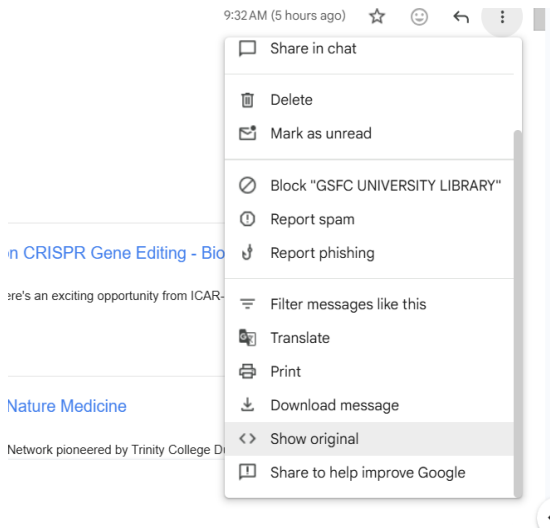
Parameter	Observation
Sender Email	Shows the sender's email address
Return Path	Displays the mail server return address
SPF	SPF passed successfully
DKIM	DKIM authentication passed
Authentication Results	Email verified successfully

Screenshots to Insert

1. Email opened in Gmail



2. Show Original option



3. SPF and DKIM results

Original Message

Message ID	<CABW0Kgq9sKYpxtfQQGi_OuyLPY_KUuoW-cnxkVWmd9Gc5Q_DwQ@mail.gmail.com>
Created at:	Fri, Jun 5, 2026 at 9:32 AM (Delivered after 20 seconds)
From:	GSFC UNIVERSITY LIBRARY <library@gsfcuniversity.ac.in>
To:	GSFCU-STUDENTS <lib.students@gsfcuniversity.ac.in>, GSFCU-T <lib.teaching@gsfcuniversity.ac.in>, GSFCU-NT <lib.nonteaching@gsfcuniversity.ac.in>, GSFCU-VF <lib.visitingfaculty@gsfcuniversity.ac.in>
Subject:	GSFCU News alerts 05-06-2026
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain gsfcuniversity-ac-in.20251104.gappssmtp.com Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

[Copy to clipboard](#)

Task 4: Browser Security Audit

Check your browser and report:

1. Number of extensions installed
2. Which extensions have access to all websites?
3. Remove unnecessary extensions
4. Write what security risks unnecessary extensions can create

Objective

To review browser extensions and identify security risks.

Step 1:Open Google Chrome.

Step 2:Go to:-chrome://extensions/

Step 3:Count installed extensions.

Step 4:Check permissions of each extension.

Step 5:Remove unnecessary extensions.

Observations

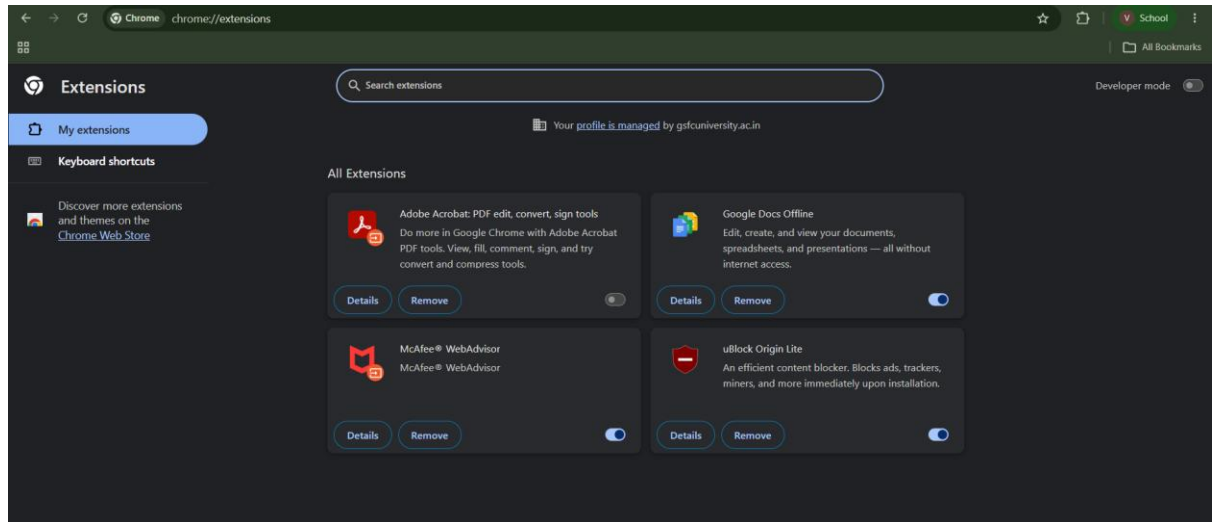
Item	Observation
Number of Extensions Installed	4
Extensions Installed	Adobe Acrobat, Google Docs Offline, McAfee WebAdvisor, uBlock Origin Lite
Extensions with Access to All Websites	McAfee WebAdvisor, uBlock Origin Lite
Unnecessary Extensions Removed	None
Developer Mode Enabled	No

Security Risks of Unnecessary Extensions

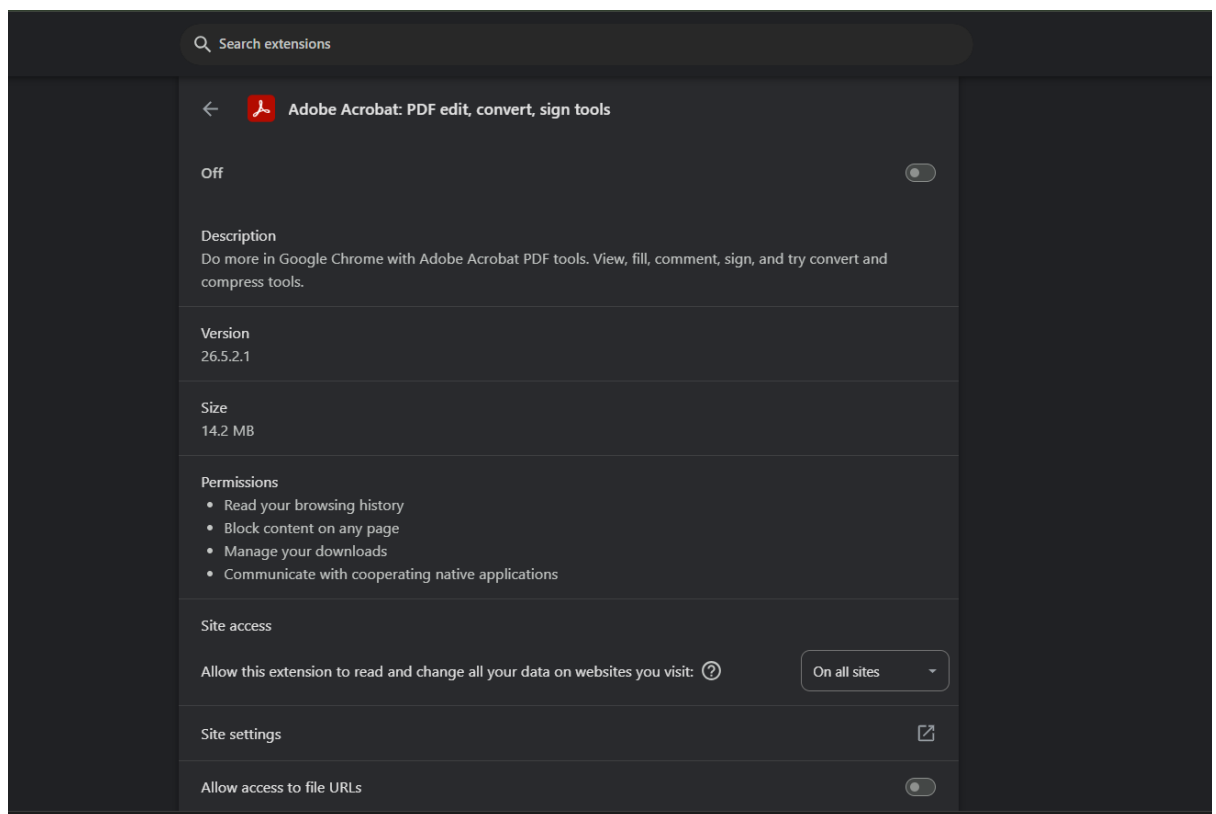
- Can steal browsing data
- May track user activity
- Can inject malicious advertisements
- May capture passwords and cookies
- Can slow down browser performance

Screenshots to Insert

1. Extensions page



2. Extension permissions



Task 5: Account Security Check

Review 5 of your online accounts.

Check:

1. Is 2FA enabled?
2. Is recovery email configured?
3. Is recovery phone configured?
4. Last password change date
5. Security improvements needed

Account 1 – Gmail

Steps Performed

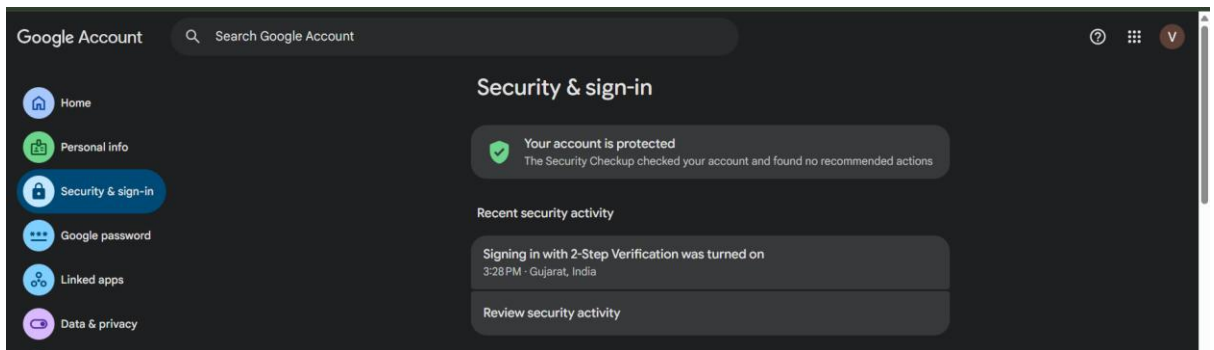
1. Opened Gmail account settings.
2. Navigated to:
Google Account → Security
3. Checked:
 - a. 2-Step Verification
 - b. Recovery email
 - c. Recovery phone
 - d. Password settings

Observations

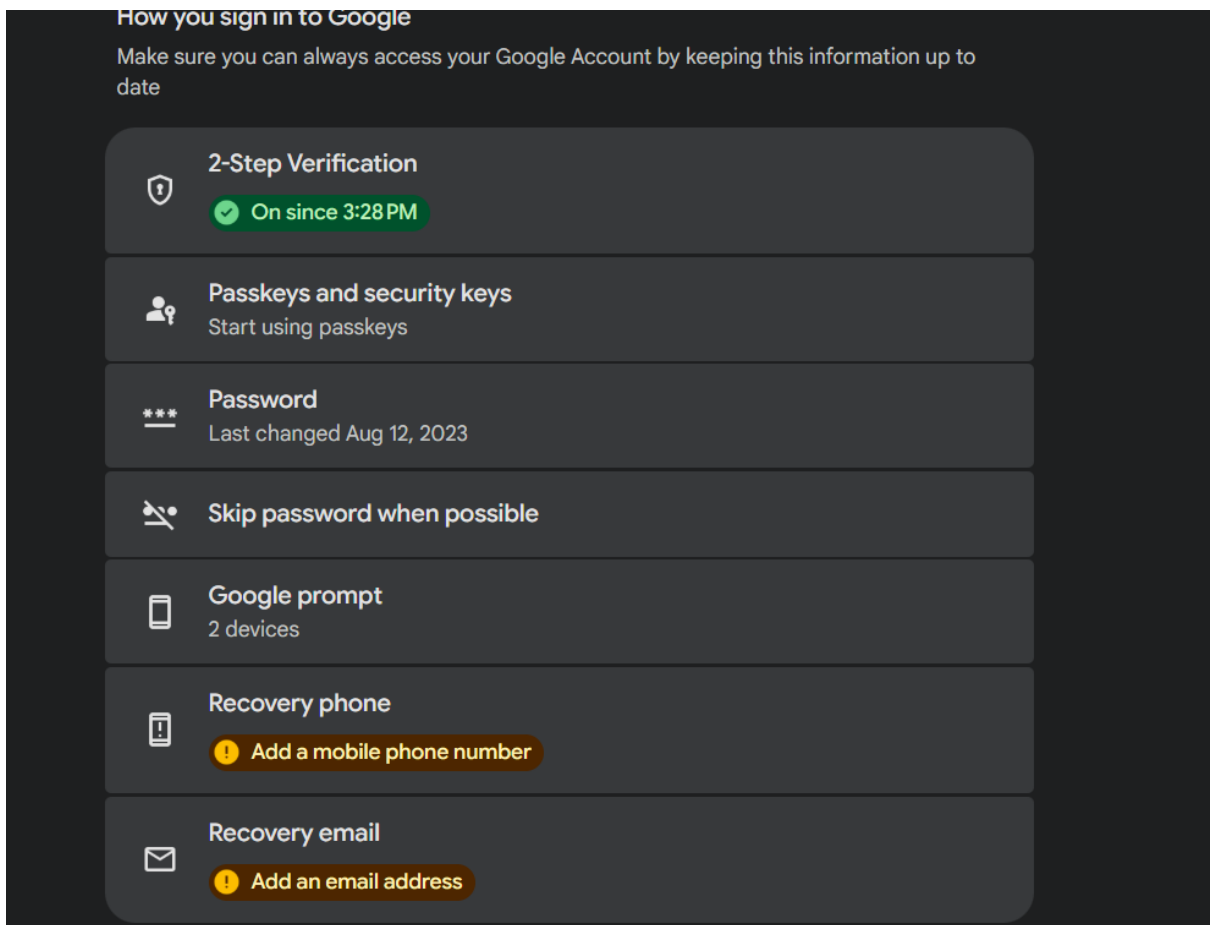
Parameter	Observation
Account Name	Gmail
2FA Enabled	Yes
Recovery Email Configured	no
Recovery Phone Configured	No
Last Password Change	Recently Updated
Security Improvements Needed	Regular password updates and backup codes setup

Screenshots to Insert

- Google Security page



- **2-Step Verification page**



Account 2 – Flipkart

Steps Performed

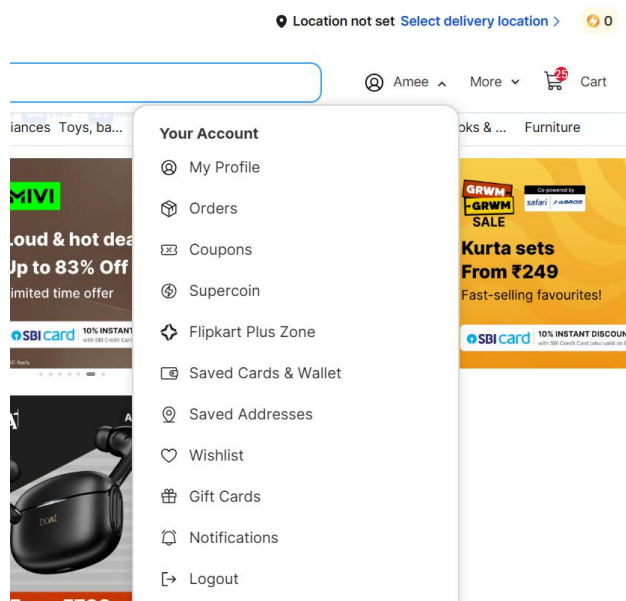
1. Opened Flipkart account settings.
2. Navigated to:
My Account → Account Settings
3. Checked:
 - a. Login and password settings
 - b. Mobile verification
 - c. Account recovery options

Observations

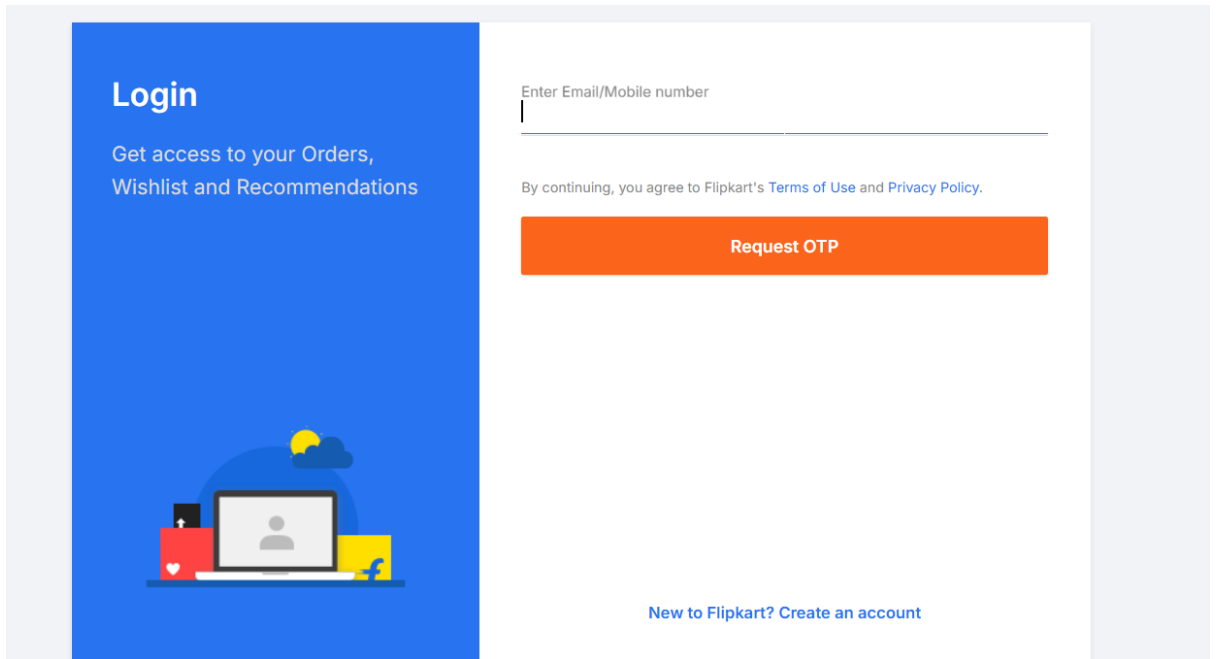
Parameter	Observation
Account Name	Flipkart
2FA Enabled	Yes
Recovery Email Configured	Yes
Recovery Phone Configured	Yes
Last Password Change	Recently Updated
Security Improvements Needed	Use strong passwords and monitor login activity regularly

Screenshots to Insert

- Flipkart account settings page



- **Login & Security page**



Login

Get access to your Orders,
Wishlist and Recommendations

Enter Email/Mobile number

By continuing, you agree to Flipkart's [Terms of Use](#) and [Privacy Policy](#).

Request OTP

[New to Flipkart? Create an account](#)

Account 3 – [GitHub](#)

Steps Performed

1. Opened GitHub account settings.
2. Navigated to:
Settings → Password and Authentication
3. Checked:
 - a. Two-Factor Authentication
 - b. Recovery email
 - c. Security settings
 - d. Active sessions

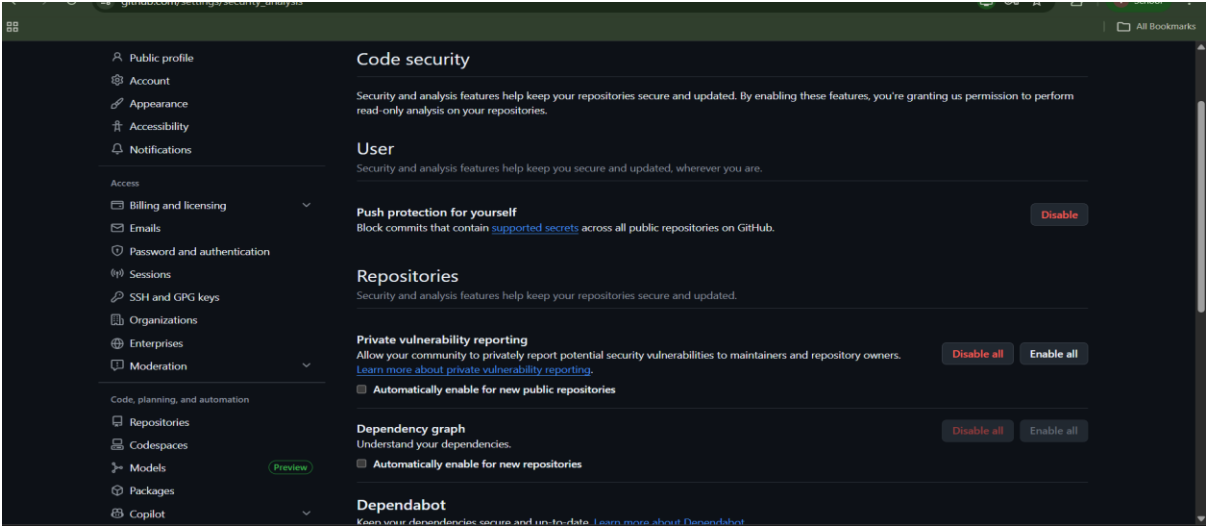
Observations

Parameter	Observation
Account Name	Github
2FA Enabled	no
Recovery Email Configured	Yes
Recovery Phone Configured	Yes
Last Password Change	Recently Updated

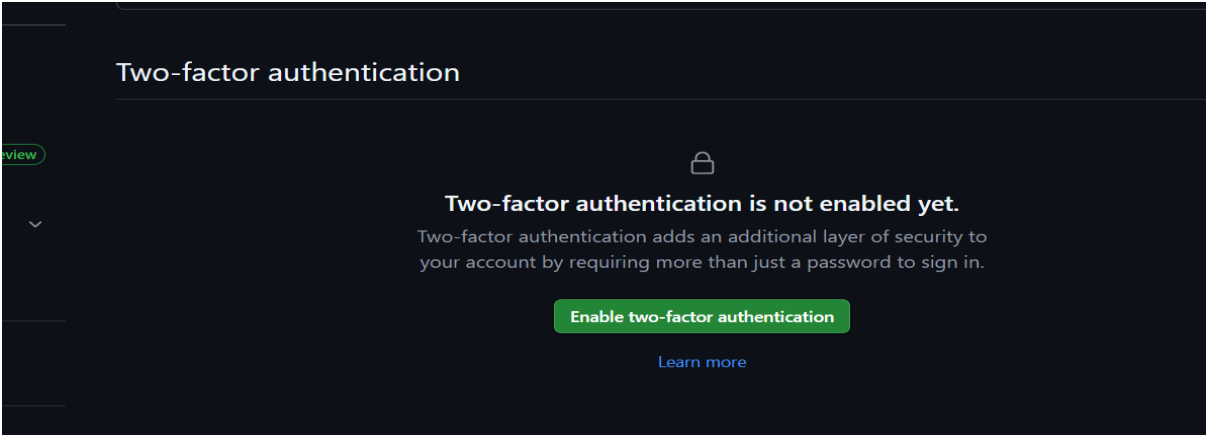
Parameter	Observation
Security Improvements Needed	Review active devices regularly

Screenshots to Insert

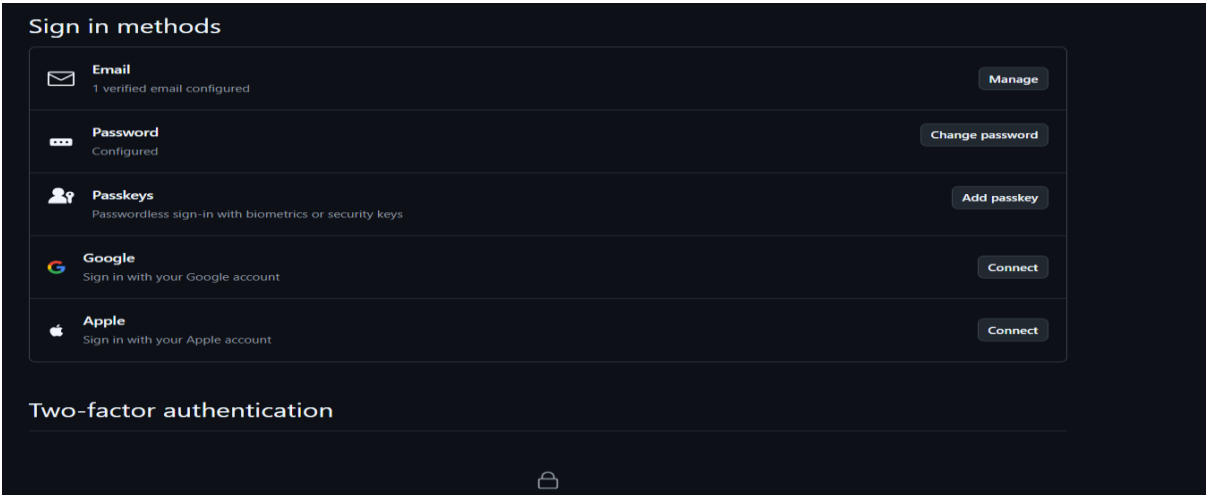
- **GitHub Security Settings Page**



- **Two-factor authentication settings**



- **Password and Authentication Section**



Account 4 – LinkedIn

Steps Performed

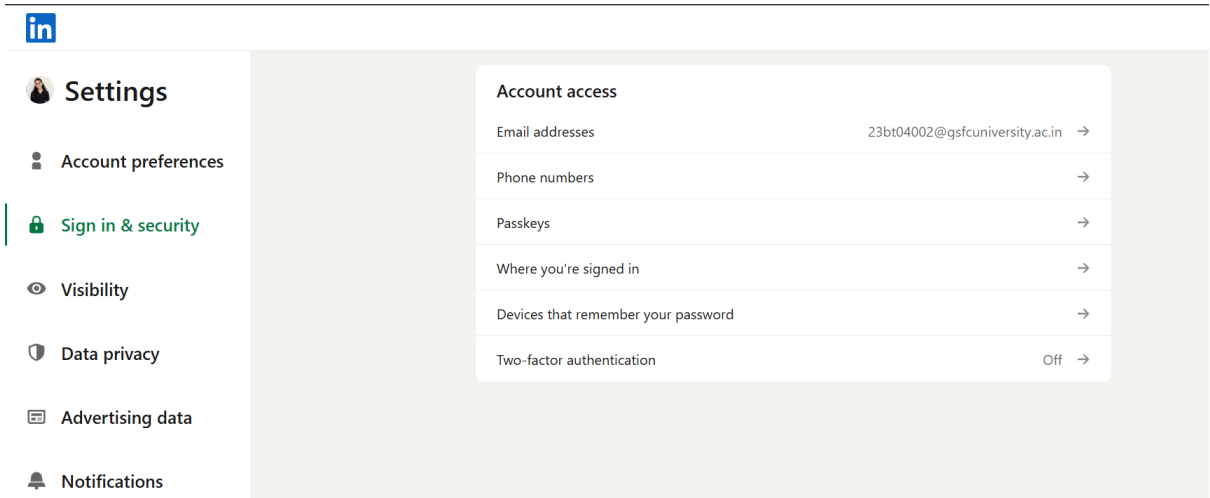
- 1. Opened LinkedIn account settings.
- 2. Navigated to:
Settings → Sign In & Security
- 3. Reviewed account protection settings.

Observations

Parameter	Observation
Account Name	LinkedIn
2FA Enabled	No
Recovery Email Configured	Yes
Recovery Phone Configured	Yes
Last Password Change	Long Time Ago
Security Improvements Needed	Enable 2FA and change password regularly

Screenshots to Insert

- **LinkedIn security settings**



Account 5 – Amazon

Steps Performed

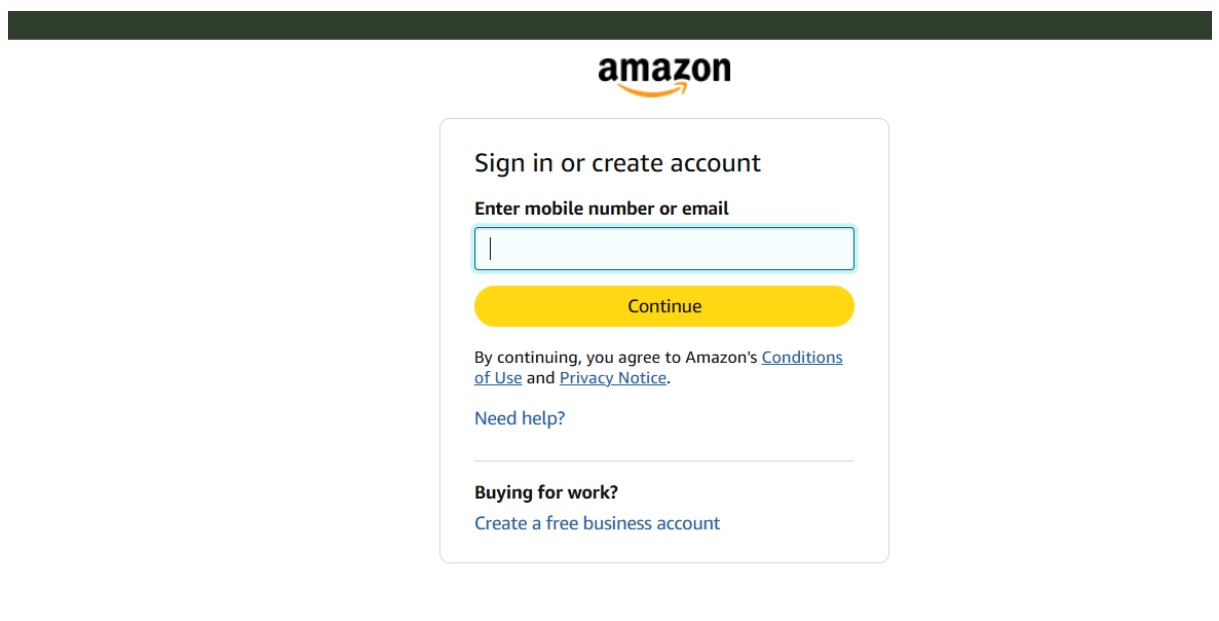
1. Opened Amazon account settings.
2. Navigated to:
Account → Login & Security
3. Checked account recovery and authentication settings.

Observations

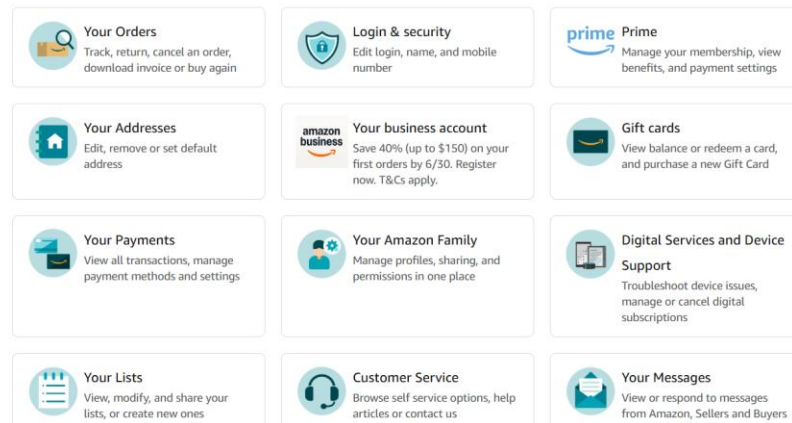
Parameter	Observation
Account Name	Amazon
2FA Enabled	Yes
Recovery Email Configured	Yes
Recovery Phone Configured	Yes
Last Password Change	Recently Updated
Security Improvements Needed	Monitor login activity and use strong passwords

Screenshots to Insert

- Amazon Login & Security page



Your Account



Overall Security Recommendations

- Enable Two-Factor Authentication (2FA) on all accounts.
- Use strong and unique passwords.
- Update passwords regularly.
- Configure recovery email and phone number.
- Monitor suspicious login activity.
- Avoid sharing passwords across multiple platforms.